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Answer **all** questions.

NEITHER MATHEMATICAL TABLES NOR SLIDE RULES NOR CALCULATORS MAY BE USED IN THIS PAPER.

1 **(a)** Express 4 560 in standard form.

Answer (a) .....

[1]

(b) It is given that $2 \times 10^3 + 1 \times 10^1 + 3 \times 10^x + 5 \times 10^y = 2\,013{,}05$
where x and y are integers.
Find the value of x and the value of y.

Answer (b) .....

[2]

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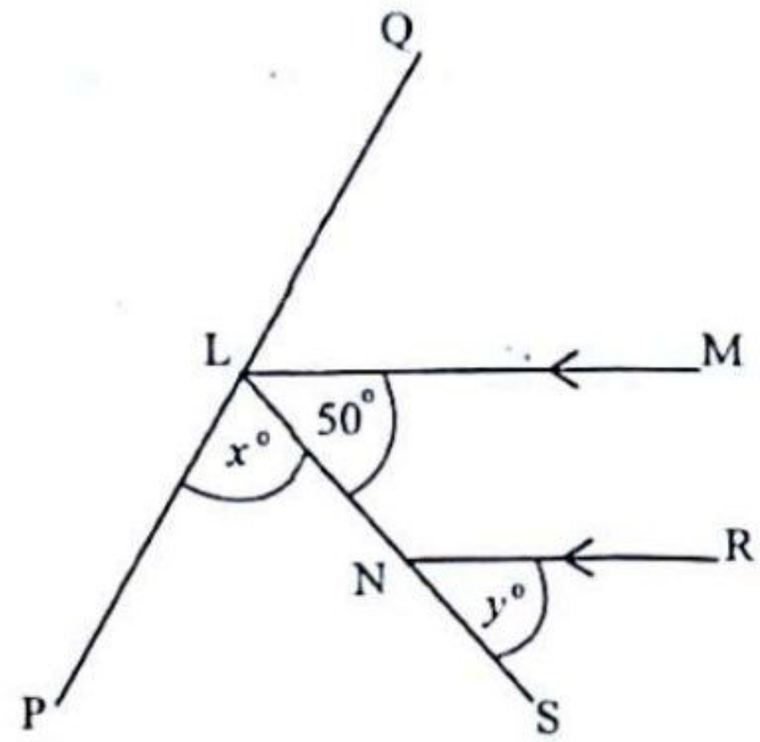
2



In the diagram, PLQ is a straight line. Line LM is parallel to line NR. The line LM bisects $Q\hat{L}N$. $N\hat{L}M = 50°$. $P\hat{L}N = x°$. $R\hat{N}S = y°$

Find the angle marked

(a) x,

Answer (a) .....

[2]

(b) y.

Answer (b) .....

[1]

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3 The temperature reading in °C, at noon, over a period of 6 days were 30, 29, x, 31, 30 and 32.

The mean temperature for the 6 days was 30,5°C.

Calculate the value of x.

[3]

4 (a) Solve the inequality $x - 8 > 4x + 1$.

Answer (a) .....

[2]

(b) Represent the solution in **(a)** on a number line.

Answer (a) .....

[1]

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5 Evaluate

(a) $0{,}2 \times 0{,}35$,

Answer (a) .....
[1]

(b) $1\frac{2}{5} \div 1\frac{2}{3}$,

Answer (b) .....
[2]

(c) $\sqrt{0,\ 0081}$.

Answer (c) .....
[1]

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6 The temperature readings at 09 00 and 16 00 were $-3°C$ and $18°C$ respectively. It is assumed that the temperature was rising at a steady rate.

Find the

(a) temperature range,

Answer (a) .....
[1]

(b) rate of temperature change per hour,

Answer (b) .....
[1]

(c) temperature at 13 00.

Answer (c) .....
[2]

7 **(a)** Round off 148,32 to the nearest ten.

Answer (a) .....
[1]

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(b) Estimate $179{,}35 \times 0{,}0496$, by expressing each of 179,35 and 0,0496. to one significant figure.

Answer (b) .....

[3]

8 A solid cuboid measures 5cm by 2cm by x cm.

(a) Form an expression, in terms of x, for the total surface area of the cuboid.

Answer (a) .....

[2]

(b) **(i)** The total surface area of the cuboid is 62cm^2.
Form an equation in x.

Answer (b)(i) .....

[1]

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(ii) Solve the equation in **(b)(i)** to find the value of x.

Answer (b)(ii) .....

[1]

9 **(a)** A solid object has a mass of 300kg and a volume of $2m^3$.
Calculate the density of the object.

Answer (a) .....

[2]

(b) A solid object has a mass of 300kg and a volume of $2m^3$.
Calculate the volume of a similar object given that its mass is 975kg.

Answer (b) .....

[2]

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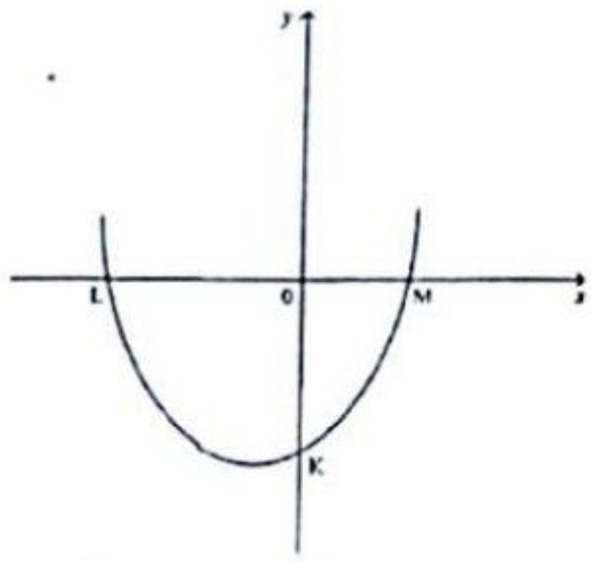
10



The diagram is a sketch of $y = x^2 + x - 12$.

The graph cuts the y - axis at (0 ; K)

It crosses the x - axis at L and M.

(a) Write down the value of K.

Answer (a) .....

[1]

(b) Find the length of LM.

Answer (b) .....

[3]

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11 y is directly proportional to the square root of x.

(a) Given that $x = 8$ when $y = 16$,
find the formula connecting x and y,

Answer (a) .....
[2]

(b) Find the value of x when $y = 12$.

Answer (b) .....
[2]

12 **(a)** Evaluate

(i) $\left(\frac{1}{3}\right)^{-2}$,

Answer (a)(i) .....
[1]

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(ii) $4^2 + 4^0$.

Answer (a)(ii) .....
[1]

(b) Solve the equation $5^{a-1} = 125$.

Answer (b) .....
[2]

13 **(a)** Factorise completely $5m^2 - 20$.

Answer (a) .....
[2]

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(b) Simplify $\dfrac{3x + 3y}{x^2 - y^2}$.

Answer (b) .....

[2]

14 The table below shows the number of children living in houses at the staff quarters at a certain school.

Number of children in a house	0	1	2	3	4	5
Number of houses	2	3	4	5	1	0

Find the

(a) mode,

Answer (a) .....

[1]

(b) median,

Answer (b) .....

[1]

(c) mean.

Answer (c) .....

[2]

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(b) Simplify $\dfrac{3x + 3y}{x^2 - y^2}$.

Answer (b) .....

[2]

14 The table below shows the number of children living in houses at the staff quarters at a certain school.

Number of children in a house	0	1	2	3	4	5
Number of houses	2	3	4	5	1	0

Find the

(a) mode,

Answer (a) .....

[1]

(b) median,

Answer (b) .....

[1]

(c) mean.

Answer (c) .....

[2]

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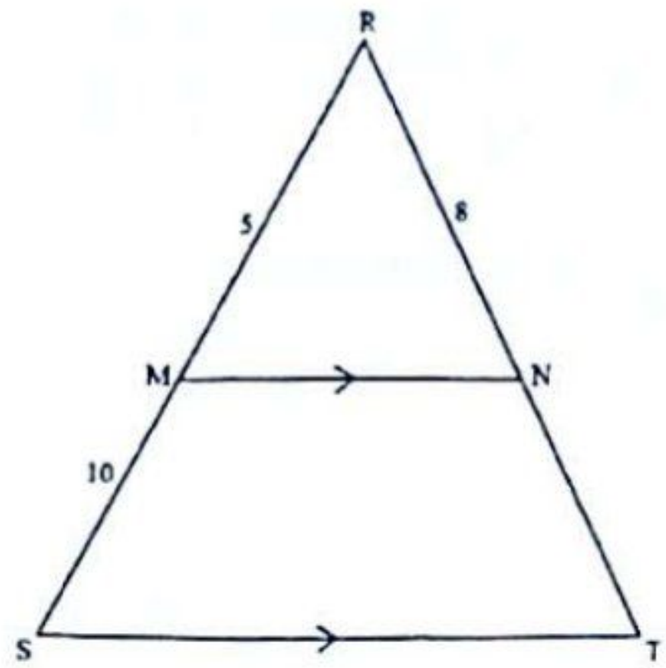
15



In the diagram is triangle RST.

MN is parallel to ST.

RM = 5 cm , $RN = 8\ cm$ and $MS = 10\ cm$

(a) Name the triangle similar to triangle RST.

Answer (a) .....

[1]

(b) Find the length of NT.

Answer (b) .....

[3]

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16 Given that $\vec{AB} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ and $\vec{BC} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$,

(a) express $\vec{AC}$ as a column vector.

Answer (a) .....
[2]

(b) find $\left|\vec{AB}\right|$.

Answer (b) .....
[2]

17 **(a)** Convert 110101_2 to a number in base five.

Answer (a) .....
[2]

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(b) Arrange the following numbers in ascending order

11010_2, 312_5 and 13_{10}

Answer (b) .....

[2]

(c) Evaluate $312_5 - 143_5$. Give the answer in base 5.

Answer (c) .....

[1]

18 The scale on a map is given as 1 : 150 000.

Find the

(a) distance on the ground represented by 1cm on the map,
Give the answer in km.

Answer (a) .....

[1]

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(b) distance on the map representing a distance on the ground of 7,5 km.
Give the answer in cm.

Answer (b) .....
[2]

(c) actual area represented by an area of 4 cm^2 on the map.
Give the answer km^2.

Answer (c) .....
[2]

19 It is given that $A = h(4m + h)$.

(a) Express m in terms of h and A,

Answer (a) .....
[3]

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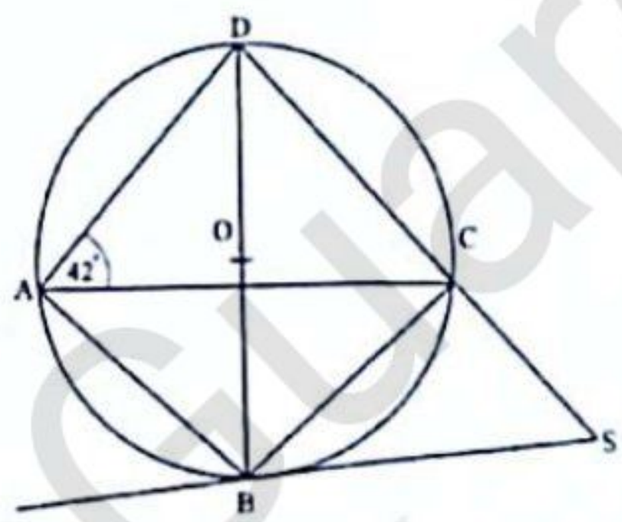
(b) Given that $A = 20$ and $h = 2$,
calculate the value of m.

Answer (b) .....
[2]

20



The diagram shows points A, B, C and D on the circumference of a circle centre O.
The line BS is a tangent to the circle at B.
DC produced meet the tangent from B at S
$D\hat{A}C = 42°$.

Find

(a) $O\hat{B}C$,

Answer (a) .....
[1]

[Turn over

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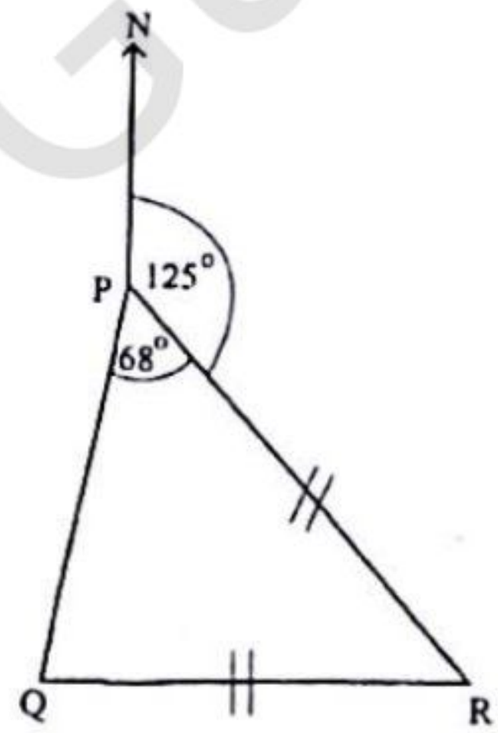
(b) $C\hat{A}B$,

Answer (b) .....
[2]

(c) $D\hat{S}B$,

Answer (c) .....
[2]

21



The diagram shows 3 towns P , Q and R on level ground.
R is equidistant from P and Q.
The bearing of R from P is $125°$.
$Q\hat{P}R = 68°$.

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Find the

(a) angle $P\hat{R}Q$,

Answer (a) .....
[2]

(b) bearing of P from R,

Answer (b) .....
[1]

(c) bearing of P from Q.

Answer (c) .....
[2]

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22 Given that $4P - 2\begin{pmatrix} 3 & 1,5 \\ -6 & 0 \end{pmatrix} = P$,

find

(a) matrix P in the form $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ where $a,\ b\ c\ and\ d$ are numbers,

Answer (a) .....

[3]

(b) the inverse of matrix P.

Answer (b) .....

[2]

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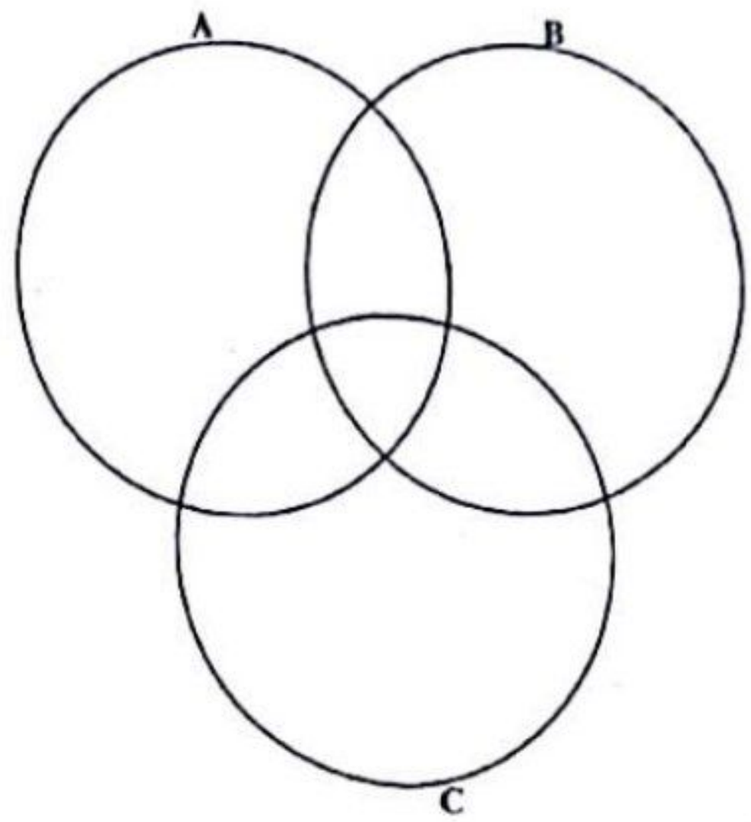
23 **(a)**



On the Venn Diagram above shade the set $(A \cap B) \cup C$.

Answer(a) .....

[2]

(b) In a class of 45 children,
30 do Agriculture.
25 do Computers.
7 do neither Agriculture nor Computers.
Find the number of children who do Agriculture only.

Answer (a) .....

[3]

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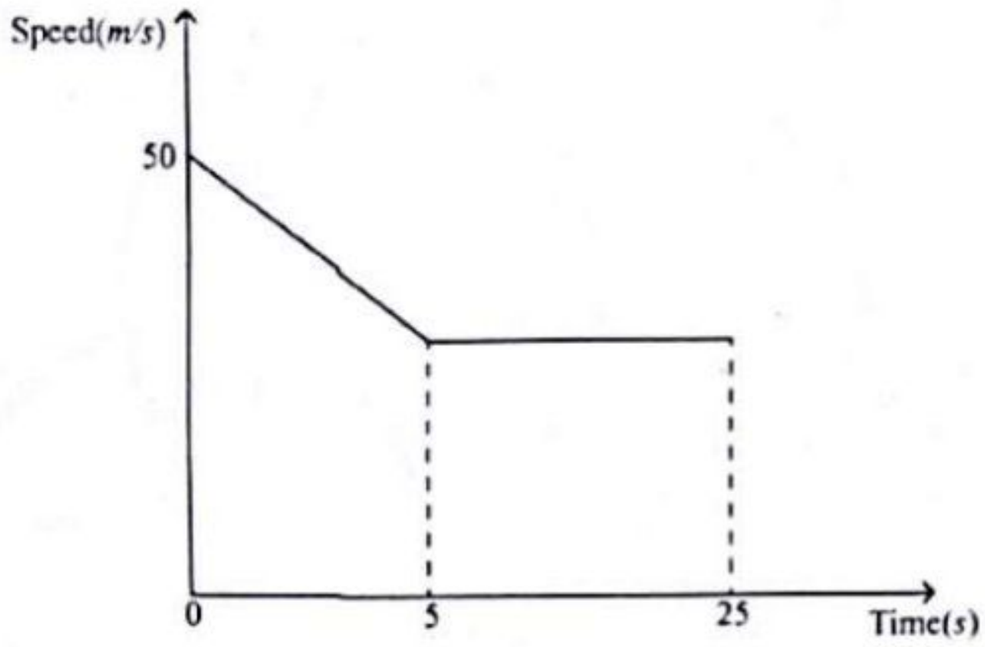
24



The diagram is a speed - time graph of a moving object.

The starting speed is 50m/s.

It decelerates at 4m/s for the first 5 seconds

It maintains constant speed for the next 20 seconds.

(a) Calculate the speed of the object when time is 10 seconds.

Answer (a) .....

[3]

(b) Calculate the distance travelled during the 25 seconds.

Answer (a) .....

[2]

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